

DeepGuard

Main Development and Design

Student	Robert B Thompson
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Document Scope: This document covers the complete design of DeepGuard: system architecture, all three AI tools in full detail, data pipeline, spacecraft system coverage, hardware integration, risk analysis, and ethical boundaries. It supports the Conceptual Design Track for ITAI 2372.

1. Design Overview

1.1 What DeepGuard Is Designed to Do

DeepGuard is an onboard AI safety monitor for deep space spacecraft. It watches every sensor on the spacecraft, detects readings that look wrong, predicts where those readings are heading, and takes a pre-approved corrective action before the situation becomes dangerous. It does all of this in under two seconds without waiting for a command from Earth.

A round-trip signal between Earth and Mars takes up to 48 minutes. In that window, a fault that starts small can become catastrophic. DeepGuard bridges that gap by giving the spacecraft the ability to protect itself.

1.2 Core Design Principles

- **Speed over perfection.** A response that is 90% correct in 2 seconds is more valuable than a perfect response in 10 minutes. DeepGuard acts fast and logs everything so humans can review and correct later.
- **Bounded authority.** DeepGuard only takes actions from a pre-approved list. It never improvises. If a situation falls outside the approved list, it alerts the crew and waits.
- **Transparency first.** Every automated action includes a complete record of what triggered it, how confident the system was, and what was done.
- **Adapt over time.** DeepGuard re-learns its definition of normal every 30 days to stay accurate as sensors age across a multi-year mission.

- **Fail safe.** If any part of DeepGuard stops working, the spacecraft reverts to its existing fixed-rule fault protection system. DeepGuard adds capability; it does not replace existing safety layers.

2. System Architecture

2.1 Four-Layer Pipeline

DeepGuard processes sensor data through four layers. Each layer takes a defined input, does one specific job, and passes a defined output to the next layer. This separation makes each layer independently testable and replaceable.

#	Name	Input	What It Does	Output	Max Time
1	Data Collection	Raw sensor signals from all 6 spacecraft systems at 10 readings per second	Cleans bad values, fills short gaps using neighboring data, scales all values to a 0-to-1 range	Clean, normalized data ready for the Sensor Watcher	5 ms
2	Sensor Watcher	Normalized sensor values from Layer 1	Scores each reading on how unusual it is; flags any reading with a high suspicion score	Suspicion score per sensor + SUSPICIOUS or OK flag	5 ms
3	Trend Forecaster	Last 48 readings per sensor (8 minutes of history)	Predicts where each sensor will be in 15 minutes using pattern-learning AI	15-minute forecast for every monitored sensor	50 ms
4	Alert and Action Engine	Suspicion score + forecast + spacecraft state	Assigns an alert level (Watch/Caution/Warning/Emergency) and runs the matching approved response	Alert level assigned, action executed, full audit log entry written	2 ms

Total pipeline time: From sensor reading arrival to completed log entry takes under 62 milliseconds. The 2-second response target builds in a large safety margin.

2.2 Data Flow Example

The table below traces a single CO₂ sensor event through all four layers.

Step	What Happens	Data Passed to the Next Layer
Layer 1	CO ₂ sensor sends a reading of 0.78 on the 0-to-1 scale. Layer 1 confirms the value is not corrupted and passes it forward.	Normalized value: 0.78
Layer 2	The Sensor Watcher compares 0.78 to the distribution of past normal CO ₂ readings and assigns a suspicion score of -0.22, which crosses the Caution threshold.	Suspicion score: -0.22 Flag: SUSPICIOUS
Layer 3	The Trend Forecaster reads the last 8 minutes of CO ₂ history and predicts the reading will reach 0.91 within 15 minutes, placing it inside the Caution zone.	Forecast: 0.91 in 15 minutes Trend: RISING

Step	What Happens	Data Passed to the Next Layer
Layer 4	Score -0.22 plus a rising forecast maps to Caution level. The engine selects Protocol L-01: increase scrubber cycle rate by 25%. The command executes and the log entry is written.	Alert: CAUTION Action: Protocol L-01 executed Audit entry written

3. Tool 1: The Sensor Watcher (Anomaly Detection)

3.1 How It Works

The Sensor Watcher uses a technique called an Isolation Forest. Think of it as randomly cutting a deck of cards until one card is left alone. Normal sensor readings cluster together and take many cuts to separate. An unusual reading sits far from the cluster and gets isolated very quickly. A reading that isolates fast receives a high suspicion score.

3.2 Why This Approach Was Chosen

Approach	How It Works	Why DeepGuard Did Not Use It
Fixed threshold rules	Triggers an alert when a value crosses a hardcoded limit	Misses slow failures; produces false alarms as sensors drift with age; cannot catch new failure types
Statistical outlier detection	Flags values more than a set number of standard deviations from the average	Assumes a bell-curve distribution of readings; spacecraft sensor data is often not bell-shaped
Neural network classifier	Trained on labeled examples of normal and faulty readings	Requires large libraries of labeled fault examples; spacecraft faults are rare and often undocumented before they occur
Isolation Forest (chosen)	Isolates unusual readings by structure without needing labeled examples	Works without prior fault examples; handles unknown failure types; fast enough for real-time onboard use

3.3 Full Design Specifications

Design Element	Specification	Reasoning
Number of decision trees	200 trees in parallel	Each tree uses a different random split sequence; combining 200 results reduces the risk of any single wrong call affecting the outcome
Expected fault rate	1% of all readings	Based on NASA's published historical fault rate for crewed spacecraft systems across ISS mission data
Alert trigger threshold	Suspicion score below -0.15	Chosen by testing against 10,000 simulated normal readings and 500 known fault readings to minimize false alarms while maintaining detection sensitivity
Feature set checked	All sensors simultaneously	Checks readings across all 6 spacecraft systems at once; allows the system to catch faults that only appear abnormal when two systems are compared together
Model update frequency	Every 30 mission days	Prevents normal sensor drift from being flagged as a fault; the model re-learns on the most recent 30 days of data so its definition of normal stays current
Inference time	Under 5 ms per reading	Leaves 57 ms of budget for the Trend Forecaster and Action Engine within the 62 ms total pipeline target
Memory footprint	Under 400 MB	Fits within the 2 GB total memory allocation on the NASA SpaceCube-3 processor alongside the other two AI tools

3.4 Handling Missing or Corrupted Sensor Data

- **Short gaps (up to 3 missing readings, 0.3 seconds):** Missing values are filled by drawing a straight line between the last good reading and the next good reading.
- **Longer gaps (more than 3 missing readings):** The last confirmed good value is held in place, a sensor-fault flag is raised, and the Sensor Watcher reduces the weight of that sensor's score until normal data resumes.

4. Tool 2: The Trend Forecaster (15-Minute Prediction)

4.1 The Problem It Solves

A sensor reading inside its normal range right now can still be heading toward a danger zone in 15 minutes. The Sensor Watcher catches readings that are already wrong. The Trend Forecaster catches readings that are about to be wrong. Together they give DeepGuard two layers of early warning.

4.2 How It Works

The Trend Forecaster uses a Long Short-Term Memory network: a type of AI designed for sequences of data that change over time, like sensor readings taken every 10 seconds. It carries a memory of what it has seen before and decides what to remember, what to forget, and what to pass forward at each step. It reads 48 data points (8 minutes) and predicts the next 90 data points (15 minutes ahead) for every sensor.

4.3 Network Architecture

Component	Specification	Plain-English Purpose
Input layer	48 timesteps, all sensors fed together	Passes 8 minutes of multi-sensor history into the network at once
First learning layer	128 memory units	Finds broad, long-range patterns such as gradual thermal drift or slow power decline
Second learning layer	64 memory units	Refines the broad patterns from the first layer into precise, short-term predictions
Output layer	One output per sensor per future timestep	Produces a 15-minute forecast for every monitored sensor simultaneously
Error function	Penalizes large errors more heavily	Ensures the forecast stays close to reality, especially for readings near danger thresholds
Training method	80% of data for training; 20% held out for testing	The held-out 20% is never seen during training; it is used only to measure final accuracy

4.4 Performance Specifications

Measurement	Target	What It Means
Average prediction error	Below 2% of full sensor range	For a sensor with a 200-unit range, predictions stay within 4 units of the actual value on average
Worst-case single error	Below 8% of full sensor range	Even the least accurate individual predictions remain useful for safety decisions
Inference time per cycle	Under 50 milliseconds	The forecast updates every 10 seconds, leaving 9,950 ms of idle time between updates
Forecast horizon	90 data points (15 minutes)	Long enough to act before a Caution reading becomes a Warning, or a Warning becomes an Emergency
Input window	48 data points (8 minutes)	Short enough to run on spacecraft hardware; long enough to capture multi-minute developing trends

5. Tool 3: The Alert and Action Engine

5.1 The Four Alert Levels

The engine receives the suspicion score and the 15-minute forecast, then works through a decision tree to assign one of four alert levels and select the matching approved response.

Level	Suspicion Score	Sensor Trend	What DeepGuard Does	How Crew Is Notified	Reversible ?
WATCH	-0.15 to -0.20	Stable or improving	Logs the event and increases monitoring frequency for this sensor	Included in the next routine status report	Yes
CAUTION	-0.20 to -0.30	Slowly getting worse	Adjusts the relevant system setting by up to 10% to slow the trend	Priority alert on crew display within 10 seconds	Yes
WARNING	-0.30 to -0.45	Rapidly getting worse	Executes the pre-approved response protocol for this specific fault type	Immediate audio and visual alert with full details shown to crew	Yes
EMERGENCY	Below -0.45	Critical level about to be reached	Starts the emergency safing sequence: powers down non-essentials, activates backups	Emergency alarm + distress beacon toward Earth	Some steps irreversible

5.2 Pre-Approved Response Protocol Library

DeepGuard can only select from protocols reviewed and approved by NASA mission engineers before launch. A sample from each spacecraft system is shown below.

ID	System	Fault Type	Response Action
T-01	Thermal	Heater drifting 15% above setpoint	Switch primary heater to backup; reduce heat load from adjacent systems by 20%
P-01	Power	Single solar panel output drops below 60% of rated	Switch affected panel loads to battery reserve; flag panel for crew inspection during next EVA
P-02	Power	Battery charge falling faster than nominal discharge curve	Reduce power draw to non-essential systems by 30%; defer next high-power activity
L-01	Life Support	CO2 scrubber efficiency drops below 85%	Increase scrubber cycle rate by 25%; alert crew to prepare the backup scrubber cartridge
L-02	Life Support	Cabin pressure declining at more than 0.1 kPa per hour	Isolate pressure zone; check seals; alert crew immediately regardless of current activity
N-01	Navigation	Primary attitude sensor disagreeing with secondary by more than 0.5 degrees	Cross-reference with star tracker; flag primary sensor as unreliable until crew confirms status
C-01	Communications	Antenna gain below threshold for 60 seconds	Switch to omnidirectional backup antenna; log pointing error for ground analysis at next contact

6. Spacecraft Systems Covered

DeepGuard monitors six spacecraft systems. The table below lists each system, its sensors, normal operating ranges, and the failure types the Sensor Watcher is trained to detect.

System	Sensors Monitored	Normal Range	Failure Types Detected
Thermal Control	Heat exchanger temperatures, heater outputs, radiator panel temperatures, fluid loop pressures	-50 to +120 °C depending on zone	Gradual heater overheating, radiator panel loss, thermal runaway in electronics bays, fluid loop pressure loss
Power Distribution	Solar panel voltage and current, battery charge level, bus voltage, circuit breaker states	Solar output: 80-100% rated; Battery: 20-95% charged	Panel output loss, battery charge anomaly, bus voltage sag, unexpected circuit breaker trips
Life Support	CO2 and O2 partial pressure, cabin total pressure, temperature and humidity, scrubber flow rates	CO2 below 0.5 kPa; O2 19.5-23.1%; Cabin pressure 101 kPa nominal	CO2 buildup, slow or fast pressure leak, oxygen depletion, humidity control failure, scrubber degradation
Navigation and Steering	Star tracker angles, gyroscope rates, reaction wheel speeds, thruster valve states	Attitude error below 0.1 degrees; Gyro drift below 0.001 deg/sec	Sensor-to-sensor disagreement, gyro drift, reaction wheel speed anomaly, unexpected thruster activity
Propulsion	Propellant tank pressures, engine chamber pressures, valve positions, flow meter readings	Tank pressure per burn plan; Chamber within 2% of rated output	Pre-ignition pressure anomaly, propellant leak, valve position mismatch, off-nominal burn profile
Communications	Antenna pointing error, signal strength, data rate, transmitter power output	Signal above mission minimum; Pointing within 0.5 degrees of target	Antenna gimbal fault, signal degradation, transmitter power drop, data rate collapse

7. Hardware Integration

7.1 NASA SpaceCube-3 Specifications vs. DeepGuard Requirements

Specification	SpaceCube-3 Value	DeepGuard Requirement	Margin / Notes
AI computing power	0.5 TFLOPS	Estimated 0.12 TFLOPS peak	58 ms response achieved vs. 62 ms budget; 4 ms margin
Power draw	15 watts total	All 3 tools: estimated 8 watts	7 watts remain for operating system and other software
Memory (RAM)	8 GB available	All 3 models: under 2 GB	6 GB available for operating system and other mission software
Storage	256 GB flash	Audit logs at 30 days: approx. 4 GB	252 GB available for other mission data and software updates

Specification	SpaceCube-3 Value	DeepGuard Requirement	Margin / Notes
Radiation tolerance	Hardened to 100 krad total dose	Mars transit: approx. 70 krad over 3 years	30 krad margin above the expected mission radiation dose
Operating temperature	-55 to +125 °C	Spacecraft thermal control holds 0-40 °C	Well within the processor's rated operating range

7.2 Software Stack

Layer	Technology	Role
Operating system	VxWorks 7 (real-time, flight-certified)	Provides the precise timing guarantees required for the 2-second response target
Sensor Watcher	scikit-learn Isolation Forest, compiled to C	Proven library; C compilation removes interpreter overhead for maximum speed
Trend Forecaster	TensorFlow Lite (embedded version)	Reduced-footprint version of TensorFlow designed specifically for embedded and edge hardware
Decision engine	Custom rule engine written in C	Written in C for maximum speed and predictability; no runtime exceptions are possible
Data logging	Structured binary log files, checksum-verified	Compact format for storage efficiency; checksum verifies log integrity even after radiation events
Ground interface	CCSDS telemetry packets over spacecraft comm system	Uses the standard NASA packet format so DeepGuard logs are readable by existing ground software

8. Risk Analysis

Risk	Likelihood	Impact	Mitigation Built Into the Design
False positive: system flags a normal reading as a fault	Medium	Low	False alarm target of below 5% is enforced in testing. Watch-level alerts are batched into routine reports, not immediate alarms, to minimize crew distraction.
False negative: system misses a real fault	Low	High	The existing fixed-threshold fault protection system remains active in parallel at all times. DeepGuard adds detection capability; it does not replace any existing safety layer.
Sensor drift causes the model to re-learn incorrect normal values	Medium	Medium	The model re-learns every 30 days but cross-validates against all other sensors. A sensor drifting independently of the other five will still be flagged as unusual.
Trend Forecaster makes an inaccurate 15-minute prediction	Medium	Low	The forecast shifts the alert level by at most one tier. A severe fault cannot be downgraded to Watch based on the forecast alone; the suspicion score is always the primary input.
Hardware fault stops DeepGuard from running	Low	Medium	DeepGuard is designed to fail gracefully. If it stops running, the spacecraft reverts to fixed-rule fault protection automatically and the crew receives an alert that DeepGuard is offline.
Training data does not fully represent Mars mission conditions	Medium	Medium	Mars rover sensor data and simulated Mars-environment data are included in the training set. Mission planners review model performance against Mars-condition test scenarios before launch approval.

9. Ethical Boundaries and Authority Limits

Boundary	What DeepGuard Is Allowed to Do	What Requires Human Confirmation
Reversible system adjustments	Adjust setpoints, switch to backup systems, increase or decrease cycle rates, redistribute power loads	No confirmation required for these actions; all are logged immediately with full context
Irreversible or high-consequence actions	Nothing; DeepGuard is blocked from these actions at the software level	Venting propellant, sealing off a module, firing thrusters, and depressurizing any compartment all require crew confirmation
Emergency safing sequence	Start the sequence, power down non-essentials, lock critical systems to safe state	Irreversible steps within the sequence require crew confirmation unless crew is determined unresponsive
Crew override	Always accepted immediately with no delay and no confirmation step required	Not applicable; crew authority is unconditional and overrides any active DeepGuard action instantly
Ground control override	Accepted whenever a communication link to Earth is available	Any ground command takes immediate precedence over DeepGuard's current action

Audit Commitment: Every action DeepGuard takes is written to a tamper-resistant log within 10 milliseconds of execution. Each entry includes: UTC timestamp, sensor readings that triggered the decision, suspicion score, forecast value, alert level, protocol ID, action description, and a checksum. NASA mission engineers review these logs after every mission phase.

10. References

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